Bridging explainability and FAIRNESS in machine learning

From measurement to deployment

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AI is everywhere now

- **Everyday & generative AI**[a]
- **Data management & search**[b]
- **Healthcare & diagnostics**[c]
- **Finance & security**[d]
- **Retail & e-commerce**[e]





[a] General-purpose assistants like ChatGPT, Gemini, Claude, etc

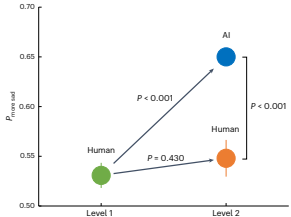
[b] Tools like Perplexity and NotebookLM turn unstructured audio, video, documents into queryable knowledge bases

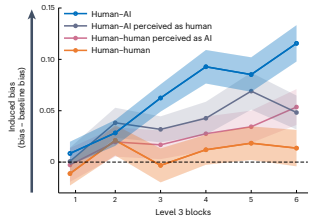
[c] AI models process medical imaging, assist in early disease detection, and accelerate pharmaceutical drug discovery

[d] ML algorithms monitor transactions in real time to flag banking fraud, automate compliance, and optimise trading

[e] Recommendation engines analyse consumer behaviour to power personalised shopping, dynamic pricing, and inventory forecasting

AI/ML is everywhere, but...





Social identity biases exist not only in human psychology and social behaviour, but also are present in artificial intelligence (AI) systems.[1]

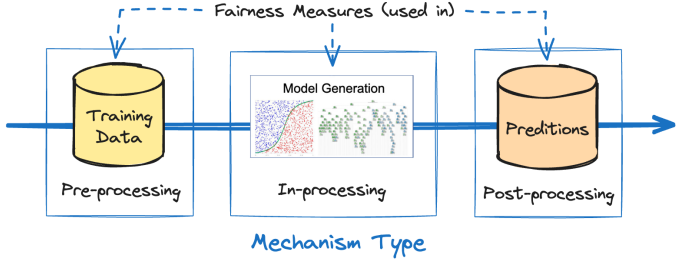
When humans and AI interact, even minute perceptual, emotional and social biases—originating either from AI systems or humans—leave human beliefs more biased, potentially forming a feedback loop.[2]

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[1] Ziad Obermeyer et al. "Dissecting racial bias in an algorithm used to manage the health of populations". In: *Science* 366.6464 (2019), pp. 447–453. DOI: 10.1126/science.aax2342; Tiancheng Hu et al. "Generative language models exhibit social identity biases". In: *Nat Comput Sci* (2024), pp. 1–11; Richard J Chen et al. "Algorithmic fairness in artificial intelligence for medicine and healthcare". In: *Nat Biomed Eng* 7.6 (2023), pp. 719–742.

[2] Moshe Glickman and Tali Sharot. "How human–AI feedback loops alter human perceptual, emotional and social judgements". In: *Nat Hum Behav* 9 (2025), pp. 345–359. DOI: 10.1038/s41562-024-02077-2; Madalina Vlasceanu and David M Amodio. "Propagation of societal gender inequality by internet search algorithms". In: *Proc Natl Acad Sci U.S.A.* 119.29 (2022), e2204529119.

Existing work about fairness

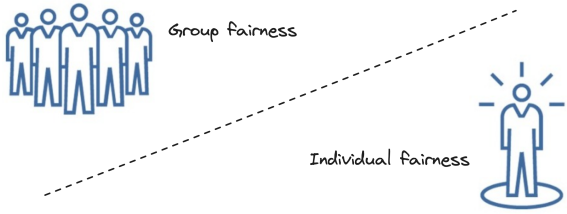


Many sources of bias [3]
Mechanisms to enhance fairness [4]

[3] *Unintential*: Limited and coarse features, sample size disparity (less data by definition about minority populations), skewed sample (feedback loops), tainted examples, features that act as proxies; *Intentional*: conscious prejudice.

[4] ***Pre-*** and ***post-processing*** *mechanisms* normally function by manipulating input or output, while ***inprocessing*** *mechanisms* introduce fairness constraints into training procedures or algorithmic objectives.

Existing work about fairness



Sources of bias, Mechanisms to enhance fairness

Types of fairness measures [3,4]

Challenging: *incompatibility,*[5] *multi-attribute protection, etc.*

[3] ***Distributive*** *fairness*: group fairness, individual fairness, counterfactual fairness, etc.; ***Procedural*** *fairness*

[4] *Group fairness* focuses on statistical/demographic equality among groups defined by sensitive attributes, while *individual fairness* follows a principle that "similar individuals should be evaluated or treated similarly."

[5] Tensions between notions of fairness, between fairness and accuracy, between different methods for achieving fairness

Table 1. Brief summary[6,7] of existing fairness metrics.[8]

Name of *measure*	Fairness type	Meaning		Applicable situation(s) in definition			Non-binary handling	
		quantitative	fairer	#label (n_c)	#sen-att (n_a)	#values per $\mathcal{A}_i$	$n_{a_i}>2$	$n_a>1$
Demographic parity (DP, aka. statistical parity)	*, group-	yes	lower value	binary	singular	bi-valued	yes	indirectly
Disparate impact /80% rule	*, group-	yes	*larger* value	binary	singular	bi-valued	yes	indirectly
Disparate treatment	*, group-	poss.	lower value	binary	singular	bi- (multi- allowed)	yes	indirectly
Conditional statistical parity	*, group-	poss.	lower value	binary	singular	multi-valued	–	indirectly
Bounded group loss	*, group-	poss.	lower value	binary	singular	multi-valued	–	indirectly
Strategic minimax fairness	*, group-	no	–	bi-/multi-	singular	multi-valued	–	indirectly
Equalised odds (EO)	*, group-	yes	lower value	binary	singular	bi-valued	yes	indirectly
Equality of opportunity (EOpp)	*, group-	yes	lower value	binary	singular	bi-valued	yes	indirectly
Predictive equality	*, group-	poss.	lower value	binary	singular	multi-valued	–	indirectly
γ-subgroup fairness	*, group-	yes	lower value	binary	singular	bi-valued	yes	indirectly
Predictive parity (PP)	*, group-	yes	lower value	binary	singular	bi-valued	yes	indirectly
Lipschitz condition	*, individual-	no	–	binary	singular	bi-valued	yes	indirectly
General entropy indices (and the Theil index)	*, individual-	yes	lower value	binary	singular	multi-valued	–	indirectly
Counterfactual fairness (CFF)	*, individual-	no	–	binary	allows plural	multi- allowed	yes	indirectly
Proxy discrimination	*, individual-	no	–	binary	singular	multi- allowed	yes	indirectly
Discriminative risk (DR)	*,[5]	yes	lower value	bi-/multi-	allows plural	multi-valued	–	–
Harmonic fairness via manifold (HFM)	*,[5]	yes	lower value	bi-/multi-	allows plural	multi-valued	–	–
Multiaccuracy	*group-	poss.	lower value	binary	singular	multi-valued	–	indirectly
Differentially fair	*group-	poss.	–	binary	allows plural	bi- (multi- allowed)	yes	indirectly
Group benefit ratio and worst-case min-max ratio	*group-	yes	*larger* value	binary	allows plural	bi- (multi- allowed)	yes	indirectly
Feature-apriori fairness	procedural	yes	–	binary	–	–	yes	yes
Feature-accuracy fairness	procedural	yes	–	binary	–	–	yes	yes
Feature-disparity fairness	procedural	yes	–	binary	–	–	yes	yes
FAE-based procedural fairness	procedural	yes	lower value	binary	singular	bi-valued	yes	indirectly

[6] Yijun Bian et al. "Algorithmic fairness: Not a purely technical but socio-technical property". In: *arXiv preprint arXiv: 2506.12556* (2025).

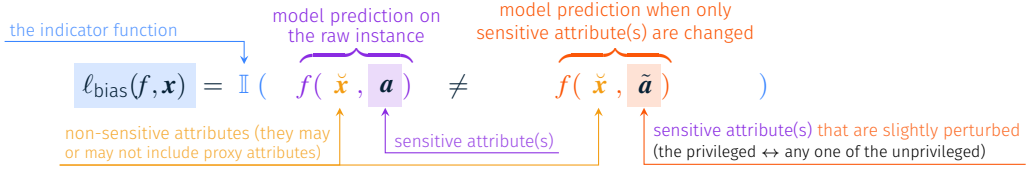
[7] Three statistical non-discrimination criteria: independence $\mathbf{A} \perp\!\!\!\perp \mathbf{R}$, separation $\mathbf{R} \perp\!\!\!\perp \mathbf{A} \mid \mathbf{Y}$, and sufficiency $\mathbf{Y} \perp\!\!\!\perp \mathbf{A} \mid \mathbf{R}$

[8] Mark* indicates it belongs to *distributive* fairness; These two can measure discrimination from both group and individual fairness aspects.

Discriminative risk (DR)[11]

—from an individual-level aspect

Following the principle of individual fairness *(the treatment/evaluation on one instance should not change solely due to minor changes in its protected/sensitive attributes)*, with an instance denoted by $\boldsymbol{x} = (\breve{\boldsymbol{x}}, \boldsymbol{a})$, the fairness quality of one hypothesis[9] $f(\cdot)$ could be evaluated by



similarly to the $0/1$ loss, where $\boldsymbol{a} = [a_1, ..., a_{n_a}]^{\mathsf{T}}$, $a_i \in \mathcal{A}_i$, $n_a \geqslant 1$, $|\mathcal{A}_i| \geqslant 2$, and $\tilde{\boldsymbol{a}}$ is a perturbed $\boldsymbol{a}$.[10] Note that Eq. (1) is evaluated on only one instance with sensitive attributes (SAs) $\boldsymbol{x}$.

[9] The hypothesis used in this equation could indicate an individual classifier or an ensemble classifier.

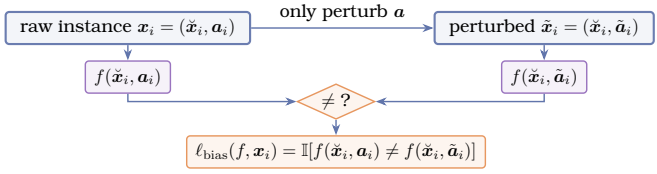
[10] A member belonging to the marginalised group(s) will be perturbed into the privileged group, while a member of the privileged group is perturbed randomly into one of the marginalised groups.

[11] Yijun Bian. "Improving fairness with ensemble combination: Margin-dependent bounds". In: *FAccT*. Montreal, QC, Canada, 2026, pp. 352–381. DOI: 10.1145/3805689.3806539.

Discriminative risk (DR) [13]

—from an individual-level aspect
—from a group-level aspect

To describe this characteristic of the hypothesis on multiple instances, then the empirical discriminative risk (DR) on a dataset S and the true DR of the hypothesis over a data distribution[12] are respectively expressed as



$$S = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$$

0 1 0 0 1 0 1

$$\hat{\boldsymbol{L}}_{\text{bias}}(f, S) = \frac{1}{n}\sum_{i=1}^{n} \ell_{\text{bias}}(f, \boldsymbol{x}_i)$$

$$\boldsymbol{L}_{\text{bias}}(f) = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}[\ell_{\text{bias}}(f, \boldsymbol{x})]$$
over population distribution

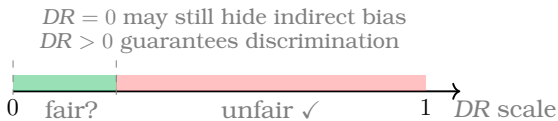
[12] The instances from S are independent identically distributed (i.i.d.) drawn from an input/feature-output/label space $\mathcal{X} \times \mathcal{Y}$ according to an unknown distribution $\mathcal{D}$.

[13] Bian, see n. 11.

Discriminative risk (DR)[12]

—from an individual-level aspect
—from a group-level aspect

To describe this characteristic of the hypothesis on multiple instances, then the empirical DR on a dataset S is expressed as $\hat{\boldsymbol{L}}_{\text{bias}}(f,S) = \frac{1}{n}\sum_{i=1}^{n} \ell_{\text{bias}}(f,\boldsymbol{x}_i)$, and the true DR of the hypothesis over a data distribution is $\boldsymbol{L}_{\text{bias}}(f) = \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}[\ \ell_{\text{bias}}(f,\boldsymbol{x})\ ]$, respectively.



Key caveat

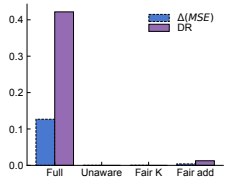
$DR = 0$ is *necessary but not sufficient* for fairness.

When DR > 0, discrimination is *guaranteed* to exist.

[12] $\hat{\boldsymbol{L}}_{\text{bias}}(f,S)$ is an *unbiased estimator* of $\boldsymbol{L}_{\text{bias}}(f)$

How DR relates to existing metrics[13,14,15]

	Group fairness[13]	Individual fairness[14]	Causal fairness[15]	*DR*
Subgroup partition	required	—	—	*not required*
Similarity metric	—	required	—	*not required*
Causal graph	—	—	required	*not required*
Multi-valued SA	✗	✗	✓	✓
Original data only	Yes	Yes	No	No



DR follows the individual-fairness principle, aggregates like group fairness, and probes sensitivity like CFF—without requiring a specific metric, a subgroup partition, or a causal model.

Limitations: *small values; computational results may be affected somehow by a randomness factor (With probability p, the instance will be perturbed into another group.)*

[13] e.g. DP, EO, EOpp, and PP. *(Empirically, DR has 0.58 Pearson correlation with accuracy change under perturbation, versus near-zero for DP & PP)*

[14] e.g. Lipschitz condition

[15] e.g. CFF, proxy discrimination. *(DR reflects CFF in a quantifiable way, with no causal graph required; a model (nearly) satisfying CFF indicates* $\mathrm{DR} \approx 0$*)*

Where DR can be useful

- Oracle bounds: can ensemble combination reduce unfairness[16]
- FairSHAP: whether and how to utilise DR to get fairer ML models[17]
- What we can study further[18]

[16] Bian, see n. 11.

[17] Lin Zhu, Yijun Bian, and Lei You. "FairSHAP: Preprocessing for fairness through attribution-based data augmentation". In: *AISTATS*. (Zhu and Bian share equal contribution). Tangier, Morocco, 2026.

[18] Bian et al., see n. 6; Yijun Bian and Yujie Luo. "Measuring model-induced discrimination via efficient fairness approximation". In: *IEEE Trans Neural Netw Learn Syst* (2026). DOI: 10.1109/TNNLS.2026.3706648; Yijun Bian and Yujie Luo. "Fast discrimination assessment for multiple non-binary sensitive attributes". In: (2026).

Oracle bounds: what we can observe

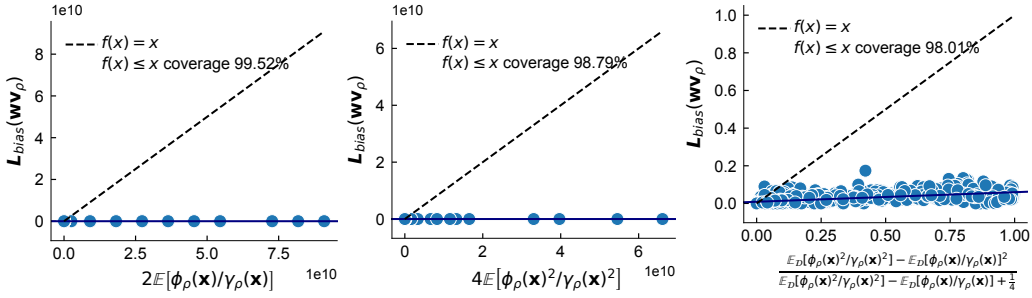


Figure 1. Verification of the proposed bounds, where (a–c) indicate theorems 3.1 to 3.3

1st order: $\boldsymbol{L}_{\text{bias}}(\mathbf{WV}_\rho) \leqslant 2\,\mathbb{E}_{\mathcal{D}}\left[\phi_\rho(\boldsymbol{x})/\gamma_\rho(\boldsymbol{x})\right]$

2nd order: $\boldsymbol{L}_{\text{bias}}(\mathbf{WV}_\rho) \leqslant 4\,\mathbb{E}_{\mathcal{D}}\left[\phi_\rho(\boldsymbol{x})^2/\gamma_\rho(\boldsymbol{x})^2\right]$

Bounds hold >98%, 2nd tighter, C-tandem tighter still

Points below the identity line = bound holds

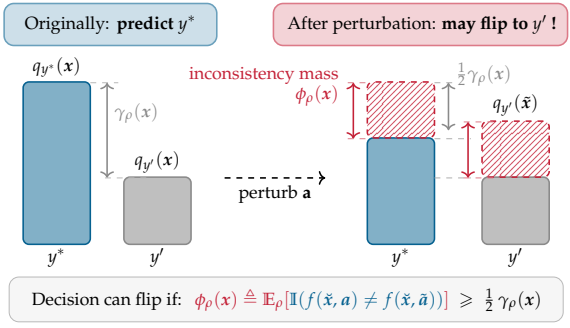
Oracle bounds:[19] what the bounds tell us

Ensembles of m classifier with weights ρ
Top-1 voting margin

$$\gamma_\rho(\mathbf{x}) \triangleq q_{y^*}(\mathbf{x}) - \max_{y \neq y^*} q_y(\mathbf{x}) \in [0, 1]$$

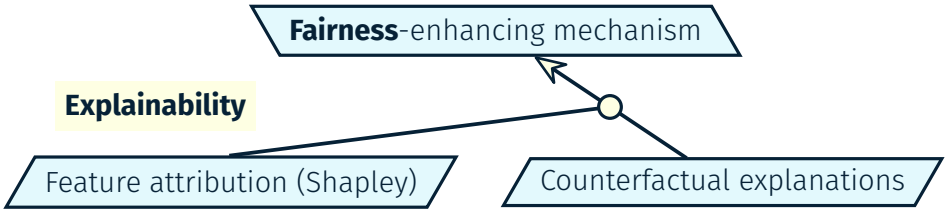
DR of ensemble $\leqslant 2\times$(individual DR/voting margin)
Larger margin $\rightarrow$ tighter bound $\rightarrow$ fairer ensemble

A *cancellation-of-biases* effect exists — but it depends on margins, unlike cancellation of errors



[19] If the ensemble exhibits discrimination risk, it means that the leading gap in the prediction class has been reversed, in other words, at least half of the votes for the prediction class will be altered. That is to say, the total weight of the inconsistency between the original prediction and the perturbed prediction has to hold

FairSHAP[20]



- Prioritising individual fairness can offer stronger and more flexible leverage than focusing only on group fairness
- Connecting explainability with fairness makes mitigation more interpretable, targeted, and practical

[20] Zhu, Bian, and You, see n. 17.

FairSHAP framework

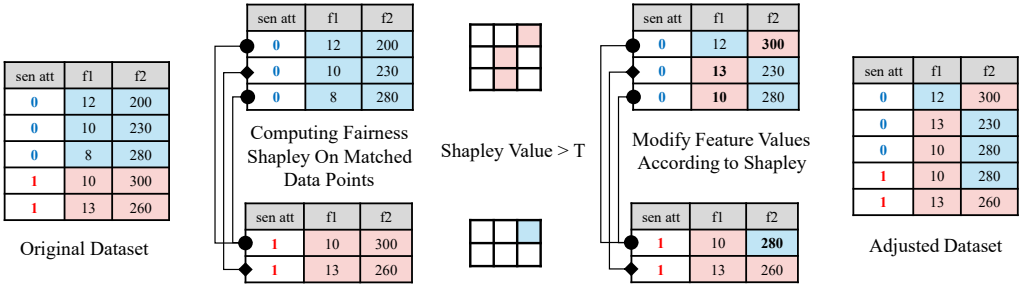


Figure 2. Overall framework of FairSHAP.[21]

· ***Left:*** *Partition the original dataset by the sensitive attribute into privileged andunprivileged groups, then perform nearest-neighbour matching between two groups.* · ***Middle:*** *Computed Shapley value matrix on the matched samples.* · ***Right:*** *Modify feature values according to the Shapley attributions and synthesize the adjusted dataset.*

[21] leverages Shapley value attribution to improve both individual and group fairness; *model-agnostic* and transparent; is broadly applicable to tabular data, supports various models and SHAP algorithms, and can be seamlessly integrated into existing ML pipelines

Qualitative results

Table 2. Compare FairSHAP with other fairness mitigation methods across different datasets.[22]
Here, "s.a." denotes the sensitive attribute used in fairness evaluation. Note that TrainingAR: Training Set Adjustment Rate; TestAN: Test Set Adjustment Necessity. Data Fidelity is measured using the Wasserstein Distance to quantify the difference between the original and adjusted data distributions.

Dataset (s.a.)	Methods	Accuracy	DR	DP	EOpp	PP	Data Fidelity	TrainingAR	TestAN
German (sex)	Baseline	0.6650±0.0257	0.0785±0.0211	0.0512±0.0346	0.1287±0.0590	0.1341±0.0486	—	—	No
	RSA	0.6590±0.0287	—	0.0522±0.0246	0.1524±0.1007	0.2189±0.0815	—	—	Yes
	CR	0.6680±0.0238	**0.0028±0.0029**	0.0844±0.0557	0.1559±0.0609	**0.0723±0.0330**	0.0183±0.0211	0.9615	Yes
	DIR	**0.6720±0.0337**	0.0966±0.0112	0.0946±0.0373	0.1737±0.0729	0.1529±0.0634	0.0155±0.0440	0.0774	Yes
	FairSHAP	0.6630±0.0275	0.0243±0.0112	**0.0301±0.0347**	**0.1126±0.0783**	0.1852±0.1074	**0.0049±0.0085**	**0.0156**	**No**
COMPAS (sex)	Baseline	**0.6698±0.0051**	0.0883±0.0064	0.1548±0.0241	0.1243±0.0510	0.0492±0.0084	—	—	No
	RSA	0.6676±0.0067	—	**0.1075±0.0160**	**0.0842±0.0475**	0.0635±0.0266	—	—	Yes
	CR	0.6679±0.0045	**0.0082±0.0070**	0.1407±0.0248	0.1291±0.0317	0.0714±0.0517	0.0189±0.0193	0.9174	Yes
	DIR	0.6644±0.0098	0.1150±0.0091	0.1155±0.0239	0.0952±0.0359	0.0747±0.0370	0.0387±0.0640	0.0650	Yes
	FairSHAP	0.6609±0.0106	0.0629±0.0091	0.1326±0.0407	0.0985±0.0603	**0.0452±0.0383**	**0.0025±0.0048**	**0.0113**	**No**
COMPAS (race)	Baseline	**0.6689±0.0108**	0.0995±0.0076	0.1436±0.0209	0.1438±0.0233	0.0522±0.0406	—	—	No
	RSA	0.6623±0.0078	—	0.2069±0.0128	0.2024±0.0349	0.0575±0.0291	—	—	Yes
	CR	0.6611±0.0112	**0.0418±0.0092**	0.1502±0.0341	0.1621±0.0530	0.0592±0.0367	0.0250±0.0222	0.8920	Yes
	DIR	0.6149±0.0286	0.1185±0.0181	0.1359±0.1241	**0.1117±0.0945**	**0.0399±0.0338**	0.0512±0.0736	0.0701	Yes
	FairSHAP	0.6627±0.0069	0.0842±0.0049	**0.1344±0.0332**	0.1568±0.0343	0.0508±0.0469	**0.0040±0.0055**	**0.0126**	**No**

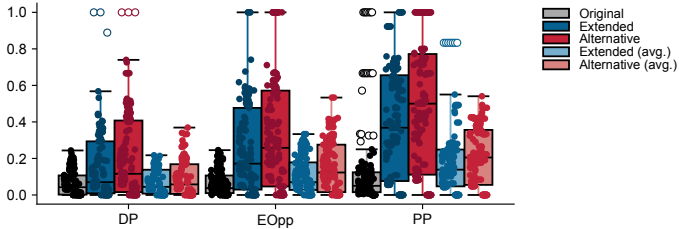
[22] "Kdd" is an abbreviation for the Census Income KDD dataset

Qualitative results

Table 2. Compare FairSHAP with other fairness mitigation methods (cont).

Dataset (s.a.)	Methods	Accuracy	DR	DP	EOpp	PP	Data Fidelity	TrainingAR	TestAN
Adult (sex)	Baseline	**0.8722**±**0.0033**	0.0315±0.0037	0.1805±0.0066	0.0735±0.0275	0.0275±0.0321	—	—	No
	RSA	0.8721±0.0024	—	0.1770±0.0066	0.0678±0.0277	**0.0272**±**0.0252**	—	—	Yes
	CR	0.8706±0.0029	**0.0000**±**0.0000**	0.1824±0.0055	0.0955±0.0243	0.0278±0.0173	0.0167±0.0391	0.9887	Yes
	DIR	0.8550±0.0067	0.0499±0.0076	0.1607±0.0157	0.0772±0.0624	0.0360±0.0253	0.0046±0.0417	0.0081	Yes
	FairSHAP	0.8692±0.0046	0.0273±0.0047	**0.1558**±**0.0130**	**0.0393**±**0.0254**	0.0474±0.0319	**0.0010**±**0.0073**	**0.0012**	**No**
Adult (race)	Baseline	**0.8721**±**0.0033**	0.0398±0.0025	0.1034±0.0110	0.0808±0.0326	0.0302±0.0265	—	—	No
	RSA	0.8710±0.0044	—	0.0971±0.0043	0.0864±0.0347	0.0427±0.0142	—	—	Yes
	CR	0.8713±0.0033	**0.0000**±**0.0000**	0.1008±0.0115	0.0983±0.0389	0.0480±0.0235	0.0300±0.0450	0.9620	Yes
	DIR	0.8320±0.0173	0.0740±0.0209	**0.0703**±**0.0515**	0.0871±0.0355	0.0482±0.0730	0.0252±0.0480	0.0089	Yes
	FairSHAP	0.8720±0.0023	0.0284±0.0017	0.0851±0.0155	**0.0287**±**0.0277**	**0.0259**±**0.0318**	**0.0030**±**0.0084**	**0.0014**	**No**
Kdd (sex)	Baseline	0.9377±0.0026	0.0767±0.0011	0.0012±0.0008	0.0012±0.0008	0.0796±0.0054	—	—	No
	RSA	0.9380±0.0026	—	**0.0008**±**0.0004**	0.0008±0.0004	0.0796±0.0055	—	—	Yes
	CR	0.9377±0.0025	**0.0000**±**0.0000**	0.0013±0.0004	0.0013±0.0003	0.0796±0.0054	0.0007±0.0013	0.9790	Yes
	DIR	0.9377±0.0028	0.0766±0.0014	0.0012±0.0010	0.0013±0.0011	0.0797±0.0055	0.0001±0.0007	0.0013	Yes
	FairSHAP	**0.9381**±**0.0029**	0.0732±0.0020	0.0008±0.0007	**0.0006**±**0.0008**	**0.0794**±**0.0060**	**0.0000**±**0.0000**	**0.0003**	**No**
DefaultCredit (sex)	Baseline	0.8141±0.0059	0.0226±0.0019	0.0335±0.0072	0.0348±0.0211	0.0269±0.0127	—	—	No
	RSA	0.8144±0.0059	—	**0.0259**±**0.0063**	0.0244±0.0122	**0.0154**±**0.0172**	—	—	Yes
	CR	0.8144±0.0068	**0.0003**±**0.0002**	0.0339±0.0077	0.0419±0.0203	0.0363±0.0203	0.0051±0.0064	0.9844	Yes
	DIR	0.8138±0.0051	0.0224±0.0027	0.0308±0.0091	0.0334±0.0244	0.0255±0.0214	0.0023±0.0056	0.0741	Yes
	FairSHAP	**0.8145**±**0.0050**	0.0214±0.0026	0.0306±0.0054	**0.0216**±**0.0164**	0.0289±0.0075	**0.0001**±**0.0003**	**0.0004**	**No**

Yet binarisation may cause problems...



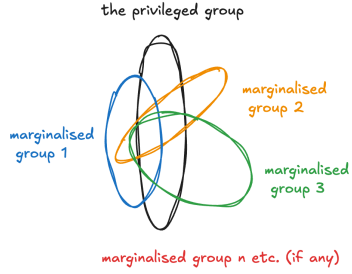
Binarisation can systematically underestimate discrimination.[22]

To bridge observational bias detection and fairness intervention...

[22] Bian et al., see n. 6.

Harmonic fairness via manifolds (HFM)[23,24]

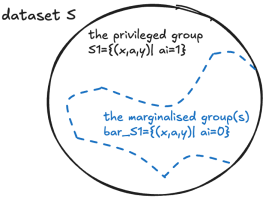
If we view the *instances (with the same value of sensitive attributes)* as *data points on certain manifold(s)*, the manifold representing members from the marginalised/unprivileged group(s) is supposed to be as close as possible to that representing members from the privileged group.



[23] Bian and Luo, "Measuring model-induced discrimination via efficient fairness approximation", see n. 18.

[24] Given a dataset $S = (\mathbf{X}, \mathbf{A}, \mathbf{Y})$, we measure fairness with respect to the SAs indicating difference from both individual- and group-apsects in three variants: (i) the *previous* HFM for a single binary SA; and (ii–iii) the *maximal (resp. average)* HFM over multiple (possibly multi-valued) SAs. HFM is built upon the concept of distances between sets, where we use the Hausdorff distance, recorded as $\mathbf{D}$, to *evaluate the discrepancy among groups divided by sensitive attributes*.

The previous HFM —for one *bi-valued* SA



Given a specific distance metric $\boldsymbol{d}(\cdot,\cdot)$[25] on the feature space, the previous HFM[26] is

$$\mathbf{df}_{\text{prev}}(f) = \mathbf{D}_f(S_1, \bar{S}_1) \big/ \mathbf{D}(S_1, \bar{S}_1) - 1\,, \tag{2}$$

where

$$\mathbf{D}.(\ S_1\ ,\ \bar{S}_1\ ;\ \ddot{y}\ ) \triangleq \max\Big\{ \max_{(\boldsymbol{x},y)\in S_1} \overbrace{\min_{(\boldsymbol{x}',y')\in \bar{S}_1} \mathbf{d}\big((\breve{\boldsymbol{x}},\ddot{y}),(\breve{\boldsymbol{x}}',\ddot{y}')\big)}^{\text{to find the nearest data point in } \bar{S}_1}\ ,\ \max_{(\boldsymbol{x}',y')\in \bar{S}_1} \overbrace{\min_{(\boldsymbol{x},y)\in S_1} \mathbf{d}\big((\breve{\boldsymbol{x}},\ddot{y}),(\breve{\boldsymbol{x}}',\ddot{y}')\big)}^{\text{to find the nearest data point in } S_1} \Big\}\,, \tag{3}$$

the privileged group ↑ S_1; marginalised/unprivileged group(s) ↑ $\bar{S}_1$; $\ddot{y}$ works for both the true label y and the prediction $\hat{y}$ of a trained classifier $f(\cdot)$

is an approximation of the distance between the manifold of unprivileged groups and that of the privileged group.[27]

[25] Here we use the standard Euclidean metric. In fact, any two metrics $\mathbf{d}_1, \mathbf{d}_2$ derived from norms on the Euclidean space $\mathbb{R}^d$ are equivalent in the sense that there are positive constants c_1, c_2 such that $c_1\mathbf{d}_1(x,y) \leqslant \mathbf{d}_2(x,y) \leqslant c_2\mathbf{d}_1(x,y)$ for all $x, y \in \mathbb{R}^d$.

[26] For a bi-valued $a_1 \in \mathcal{A}_1 = \{0,1\}$, S is divided into $S_1 = \{(\boldsymbol{x},y) \triangleq (\breve{\boldsymbol{x}}, a_1, y) \in D \mid a_1 = 1\}$ and $\bar{S}_1 = S \setminus S_1$.

[27] Notations $\mathbf{D}_f(S_1,\bar{S}_1) = \mathbf{D}.(S_1,\bar{S}_1;f(\breve{\boldsymbol{x}},a_1))$, $\mathbf{D}(S_1,\bar{S}_1) = \mathbf{D}.(S_1,\bar{S}_1;y)$ are two abbreviations for brevity.

HFM —over multiple (possibly multi-valued) SAs

For one or more multi-valued SAs $\boldsymbol{a} \in \mathcal{A}$ [25], the maximal (resp. average) HFM[26] are

$$\mathbf{df} = \log\left(\mathbf{D}_{f_{\boldsymbol{a}}}(S) \,/\, \mathbf{D}_{\boldsymbol{a}}(S)\right) , \tag{2a}$$

$$\mathbf{df}^{\mathsf{avg}}(f) = \log\left(\mathbf{D}^{\mathsf{avg}}_{f_{\boldsymbol{a}}}(S) \,/\, \mathbf{D}^{\mathsf{avg}}_{\boldsymbol{a}}(S)\right) , \tag{2b}$$

where

$$\mathbf{D}_{\cdot,\boldsymbol{a}}(S;\ddot{\mathbf{y}}) = \max_{1 \leqslant i \leqslant n_a} \mathbf{D}_{\cdot,\boldsymbol{a}}(S, a_i;\ddot{\mathbf{y}}) , \tag{3a}$$

$$\mathbf{D}^{\mathsf{avg}}_{\cdot,\boldsymbol{a}}(S;\ddot{\mathbf{y}}) = \tfrac{1}{n_a}\textstyle\sum_{i=1}^{n_a} \mathbf{D}^{\mathsf{avg}}_{\cdot,\boldsymbol{a}}(S, a_i;\ddot{\mathbf{y}}) , \tag{3b}$$

to fine the nearest data point in $\bar{S}_j$

$$\mathbf{D}_{\cdot,\boldsymbol{a}}(S, a_i;\ddot{\mathbf{y}}) = \max_{i \in [n_{a_i}]}\{\max_{(\boldsymbol{x},y)\in S_j} \overbrace{\min_{(\boldsymbol{x}',y')\in\bar{S}_j} \boldsymbol{d}((\check{\mathbf{x}},\ddot{\mathbf{y}}),(\check{\mathbf{x}}',\ddot{\mathbf{y}}'))}\}$$

$$\mathbf{D}^{\mathsf{avg}}_{\cdot,\boldsymbol{a}}(S, a_i;\ddot{\mathbf{y}}) = \tfrac{1}{n}\textstyle\sum_{j\in[n_{a_i}]}\sum_{(\check{\mathbf{x}},y)\in S_j} \min_{(\check{\mathbf{x}}',y')\in\bar{S}_j} \boldsymbol{d}((\check{\mathbf{x}},\ddot{\mathbf{y}}),(\check{\mathbf{x}}',\ddot{\mathbf{y}}')) .$$

[25] where $n_a \geqslant 1$ and $|\mathcal{A}_i| \geqslant 2\ (i \in [n_a])$

[26] Note that $S_j = \{(\boldsymbol{x}, y) \in S \mid a_i = j\}$, $\bar{S}_j = S \setminus S_j$, and special case $\mathbf{D}_{\cdot,\boldsymbol{a}}(S, a_i;\ddot{\mathbf{y}}) = \mathbf{D}_{\cdot}(S_1, \bar{S}_1;\ddot{\mathbf{y}})$ when $\mathcal{A}_i = \{0, 1\}$.

Approximating HFM[28]

	Work 1	Work 2	
	previous	maximal	average
Distance between sets	$\mathbf{D}(S_1, \bar{S}_1; \ddot{y})$	$\mathbf{D}_{\cdot,\boldsymbol{a}}(S, a_i)$	$\mathbf{D}^{\text{avg}}_{\cdot,\boldsymbol{a}}(S, a_i)$
		$\mathbf{D}_{\cdot,\boldsymbol{a}}(S)$	$\mathbf{D}^{\text{avg}}_{\cdot,\boldsymbol{a}}(S)$
HFM (fairness measure)	$\mathbf{df}_{\text{prev}}(f)$	$\mathbf{df}(f)$	$\mathbf{df}^{\text{avg}}(f)$
Approximation for one SA	AcceleDist		
	ApproxDist		StratES, StratRA[27]
Approximation for several SAs			EffExact

[27] and a revised ApproxDist

[28] Bian and Luo, "Measuring model-induced discrimination via efficient fairness approximation", see n. 18; Bian and Luo, "Fast discrimination assessment for multiple non-binary sensitive attributes", see n. 18.

Appendix

- Verifying margin-dependent bounds
- FairSHAP preliminaries

Margin-dependent bounds[29]

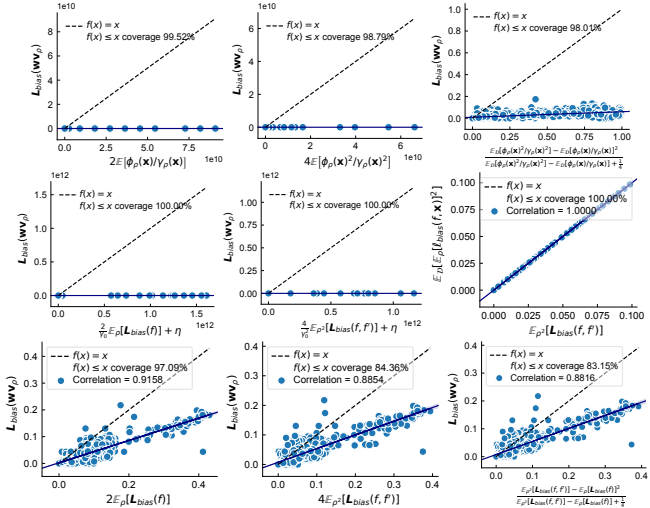


Figure 3. Verification of the proposed margin-dependent bounds and their relaxation

Margin-dependent bounds[29]

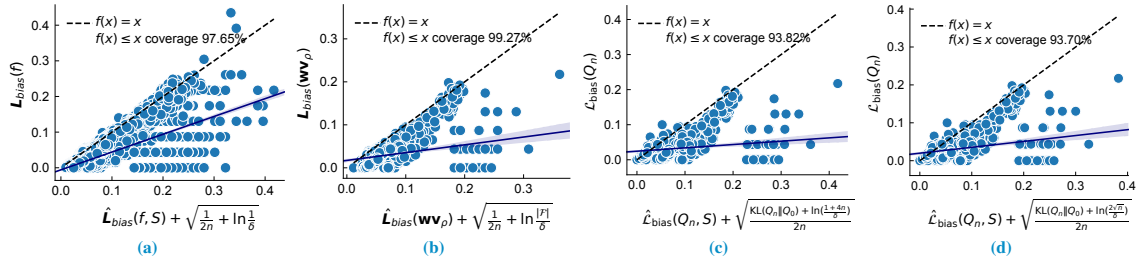


Figure 3. Verification of the proposed PAC and PAC-Bayes bounds

[29] Bian, see n. 11.

FairSHAP prelim[30]

Shapley value

For a set of players/elements $\mathcal{F} = \{1, 2, \ldots, n\}$ and a characteristic function $v : 2^{\mathcal{F}} \to \mathbb{R}$ that assigns a value to each coalition/combination $S \subseteq \mathcal{F}$, the Shapley value of player k is defined as

$$\phi_k(v) = \sum_{S \subseteq \mathcal{F} \setminus \{k\}} \frac{|S|!(n-|S|-1)!}{n!} \Big(v(S \cup \{k\}) - v(S) \Big). \tag{4}$$

Discriminative risk

With $\mathbf{x} = (\check{\mathbf{x}}, A)$, where $\check{\mathbf{x}} \in \mathcal{X}_{\mathcal{F} \setminus \{A\}}$ are the non-sensitive features and $A \in \{0, 1\}$ is the sensitive attribute (0 = unprivileged, 1 = privileged), DR is $L_{\text{DR}}(f, \mathbf{x}) = \big| f(\check{\mathbf{x}}, A = 0) - f(\check{\mathbf{x}}, A = 1) \big| \leq \varepsilon$.

FairSHAP

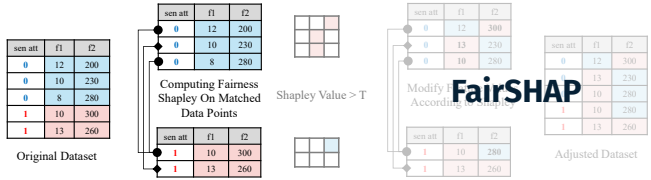
For instance i, define the coalition value on features $S \subseteq \mathcal{F}$ via an individual-fairness loss, where $\mathcal{P}$ is the joint probability obtained by nearest neighbour matching:

$$v^{(i)}(S) = \mathbb{E}_{\tilde{\boldsymbol{g}} \sim \mathcal{P}(\tilde{\boldsymbol{g}} | \boldsymbol{g}_i)} \left[L_{\text{DR}} \left(\boldsymbol{g}_{i,S}; \tilde{\boldsymbol{g}}_{\mathcal{F} \setminus S} \right) \right] - \mathbb{E}_{\tilde{\boldsymbol{g}} \sim \mathcal{P}(\tilde{\boldsymbol{g}} | \boldsymbol{g}_i)} \left[L_{\text{DR}} \left(\tilde{\boldsymbol{g}}_{\mathcal{F}} \right) \right], \tag{5a}$$

$$\text{s.t.} \quad \mathcal{P}(\boldsymbol{g}, \tilde{\boldsymbol{g}}) = \mathcal{M}_{\text{method}}(\mathcal{G}, \tilde{\mathcal{G}}). \tag{5b}$$

[30] Zhu, Bian, and You, see n. 17.

FairSHAP: algorithm 1



Algorithm 1. Overall framework: Enhancing fairness via matching and Shapley values

Input: Model f, dataset $\mathcal{D} \in \mathbb{R}^{(n+m)\times d}$ with sensitive attribute $A \in \{0,1\}$, threshold T, matching method $\mathcal{M}_{\text{method}}$, where method $\in \{\text{NearestNeighbour}, \text{OptimalTransport}\}$

Output: $\mathcal{D}_{\text{new}}$

1: Split $\mathcal{D}$ into two subgroups: $\mathcal{G} \in \mathbb{R}^{n\times d}$ (e.g., $A = 0$) and $\widetilde{\mathcal{G}} \in \mathbb{R}^{m\times d}$ (e.g., $A = 1$)
2: $\mathcal{G}' \leftarrow$ **FairSHAP**(target $= \mathcal{G}$, non-target $= \widetilde{\mathcal{G}}$, model $= f$, T, $\mathcal{M}_{\text{method}}$) // see Algorithm 2
3: $\widetilde{\mathcal{G}}' \leftarrow$ **FairSHAP**(target $= \widetilde{\mathcal{G}}$, non-target $= \mathcal{G}$, model $= f$, T, $\mathcal{M}_{\text{method}}$) // see Algorithm 2
4: $\mathcal{D}_{\text{new}} \leftarrow \text{Concat}(\mathcal{G}', \widetilde{\mathcal{G}}') \in \mathbb{R}^{(n+m)\times d}$
5: **return** $\mathcal{D}_{\text{new}}$

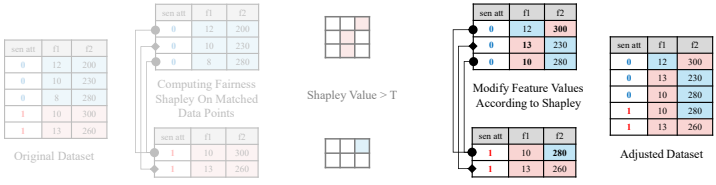
FairSHAP: algorithm 2

Algorithm 2. `FairSHAP`

Input: Target group $\mathcal{G} \in \mathbb{R}^{n\times d}$ and non-target group $\widetilde{\mathcal{G}} \in \mathbb{R}^{m\times d}$, model f, threshold T, matching method $\mathcal{M}_{\text{method}}$

Output: modified dataset $\mathcal{G}'$

Use $\mathcal{M}_{\text{method}}(\mathcal{G}, \widetilde{\mathcal{G}})$ to obtain joint probability $\mathcal{P}(\mathbf{g}, \tilde{\mathbf{g}}) \in \mathbb{R}^{n\times m}$
Use Eqs. (4) and (5) to obtain Shapley value matrix $\boldsymbol{\phi} \in \mathbb{R}^{n\times d}$
Initialize reference data $\mathcal{B} \leftarrow \mathbf{0}_{n\times d}$
for $i = 1$ **to** n **do**
  $j^* \leftarrow \arg\max_{1\le j\le m} \mathcal{P}_{i,j}$
  $\mathcal{B}_{i,:} \leftarrow \widetilde{\mathcal{G}}_{j^*,:}$
Let $\mathcal{G}' \leftarrow \mathcal{G}(\mathcal{G}' \in \mathbb{R}^{n\times d})$
for $i = 1$ **to** n **do**
  for $k = 1$ **to** d **do**
    if $\phi_{i,k} \ge T$ **then**
      $\mathcal{G}'_{i,k} \leftarrow \mathcal{B}_{i,k}$
return $\mathcal{G}'$



Qualitative results

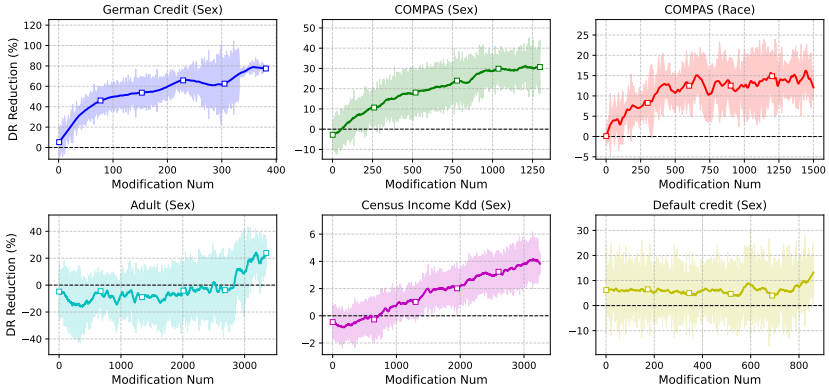


Figure 4. Percentage reduction in the DR across different datasets. *The x-axis denotes the number of modifications applied (up to the maximum required under a fairness threshold $T=0.05$), while the y-axis indicates the relative in DR, expressed as a percentage of the original value.*

Qualitative results

Table 3. Compare FairSHAP with ablation studies across different datasets. *The sensitive attribute is sex in all cases. Best and second-best results for fairness measures (DR, DP, EOpp, PP) are **bold**.*

Dataset	Methods	Accuracy	DR	DP	EOpp	PP
German	Baseline	0.6650±0.0257	0.0785±0.0211	0.0512±0.0346	0.1287±0.0590	**0.1341±0.0486**
	Ablation study 1	**0.6690±0.0237**	0.0709±0.0239	0.0446±0.0106	0.0972±0.1091	0.1463±0.1133
	Ablation study 2	0.6470±0.0452	0.0640±0.0176	0.0708±0.0510	0.1619±0.0730	0.1475±0.1384
	FairSHAP	0.6630±0.0275	**0.0243±0.0112**	**0.0301±0.0347**	**0.1126±0.0783**	0.1852±0.1074
COMPAS	Baseline	0.6698±0.0051	0.0883±0.0064	0.1548±0.0241	0.1243±0.0510	0.0492±0.0084
	Ablation study 1	**0.6713±0.0090**	0.0903±0.0110	0.1519±0.0114	0.1315±0.0379	0.0490±0.0465
	Ablation study 2	0.6699±0.0076	0.0721±0.0059	0.1537±0.0177	0.1125±0.0431	**0.0318±0.0121**
	FairSHAP	0.6609±0.0106	**0.0629±0.0091**	**0.1326±0.0407**	**0.0985±0.0603**	0.0452±0.0383
Adult	Baseline	0.8722±0.0033	0.0315±0.0037	0.1805±0.0066	0.0735±0.0275	**0.0275±0.0321**
	Ablation study 1	0.8717±0.0031	0.0332±0.0020	0.1842±0.0048	0.0932±0.0186	0.0344±0.0228
	Ablation study 2	**0.8722±0.0016**	0.0278±0.0018	0.1816±0.0084	0.0761±0.0325	0.0442±0.0266
	FairSHAP	0.8692±0.0046	**0.0273±0.0047**	**0.1558±0.0130**	**0.0393±0.0254**	0.0474±0.0319
CensusIncome	Baseline	0.9377±0.0026	0.0767±0.0011	0.0012±0.0008	0.0012±0.0008	0.0796±0.0054
	Ablation study 1	0.9377±0.0027	0.0765±0.0019	0.0010±0.0006	0.0011±0.0007	0.0797±0.0055
	Ablation study 2	0.9378±0.0024	0.0744±0.0018	0.0010±0.0003	0.0010±0.0003	0.0796±0.0056
	FairSHAP	**0.9381±0.0029**	**0.0732±0.0020**	**0.0008±0.0007**	**0.0006±0.0008**	**0.0794±0.0060**
DefaultCredit	Baseline	0.8141±0.0059	0.0226±0.0019	0.0335±0.0072	0.0348±0.0211	**0.0269±0.0127**
	Ablation study 1	0.8149±0.0061	0.0221±0.0027	0.0327±0.0067	0.0431±0.0111	0.0369±0.0166
	Ablation study 2	**0.8150±0.0051**	0.0217±0.0031	0.0331±0.0070	0.0259±0.0135	0.0325±0.0137
	FairSHAP	0.8145±0.0050	**0.0214±0.0026**	**0.0306±0.0054**	**0.0216±0.0164**	0.0289±0.0075